Agam Rain Cohen

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EDUCATION

The University of Edinburgh (Final Year) [First Class] Sept 2023 – May 2027

Course: Computer Science (BSc Hons)

Key Modules: Algorithms and Data Structures, Data Science, Computer Systems, Software Engineering, Operating Systems, Computer Security, Computer Communications and Networks, Distributed Systems

WORK EXPERIENCE

Bright Network, Technology & Engineering Intern June 2026

- Completed a Sector Skills Project, developing a Python telemetry processing script for wind turbine monitoring.
- Configured a Docker container and designed an AWS cloud architecture diagram to support the deployment pipeline.
- Authored a comprehensive engineering summary report detailing the system architecture and data processing logic.

The University of Edinburgh - School of Informatics, Undergraduate Researcher Nov 2024 – May 2025

- Co-authored a research poster and paper on the Academic Families project, investigating the role of peer-support networks in student wellbeing and community-building within computing education.
- Applied qualitative data analysis by thematically coding and visualising insights from 7 focus groups, generating evidence-based recommendations for enhancing student experience.
- Streamlined research workflows and ensured ethical compliance by securely managing participant data and consent forms.

The University of Edinburgh - School of Informatics, Technical Communications Intern Oct 2024 – May 2025

- Analysed engagement metrics by programmatically collecting platform data; these insights informed a new content strategy that increased user engagement by 40%.
- Designed and executed data-driven digital campaigns for 10+ student events, implementing A/B testing on promotional materials to optimise for higher conversion rates and boost attendance.

PROJECT EXPERIENCE

Custom Memory Allocator (C++17) github.com/agamlaugh/custom-memory-allocator

- Built a heap allocator from scratch on `mmap`/`munmap`, replacing `malloc`/`free` with slab allocation across 8 size classes (8 to 1024 bytes). Pages are dedicated to a single block size, which gives constant-time allocation and avoids external fragmentation.
- Stored free-list pointers inside the free blocks themselves (an intrusive list), so slab blocks carry no per-block header, and carved 4KB slabs lazily from a 1MB pool.
- Added a large-object path (over 1024 bytes) that calls `mmap` directly with a size header, plus 8-byte alignment and exception-based error handling (`std::invalid_argument`, `std::bad_alloc`).
- Developed test-first with 9 tests, including a 1,000-allocation stress test. The out-of-memory test exposed an integer overflow in the large-object path, which I fixed with a wrap-around check.
- Benchmarked against `malloc`/`free` and `new`/`delete` with Google Benchmark: 3–6× faster on small objects (167M vs. 27.5M objects/s on a mixed-size workload), and about 100× slower on 4KB objects, where each request makes its own `mmap` call.

Peer-to-Peer File-Sharing Node (C++17, Asio) – Ongoing github.com/agamlaugh/p2p-node

- Designed a length-prefixed TCP protocol with 14 message types, building big-endian headers with explicit bit-shifts so framing is independent of platform and host endianness.
- Implemented a stream-reassembly buffer handling TCP fragmentation, coalesced packets and split headers, decoding messages into a single event-callback interface.
- Hardened the parser against malicious peers: handshake-first enforcement, payload and buffer size limits, and bounds checks on malformed messages.
- Wrote 35+ test cases test-first with Google Test, running under AddressSanitizer.

LEADERSHIP

University of Edinburgh CompSoc, Committee Member June 2024 – May 2025

- Organised and facilitated technical workshops, networking events, and social activities for computer science students.
- Managed society logistics and outreach to enhance community engagement and student participation.

School of Informatics, Academic Parent Sept 2024 – June 2025

- Mentored a group of first-year Computer Science students, providing technical guidance on coursework and advising on effective time management and study strategies.

SKILLS

Programming Languages: C++, Python, Java, Bash, C, Haskell

Tools & Technologies: GitHub, VSCode, Linux, IntelliJ, Docker, Maven, Spring Boot, POSIX mmap/munmap, g++, Asio, Google Test, CMake, AddressSanitizer